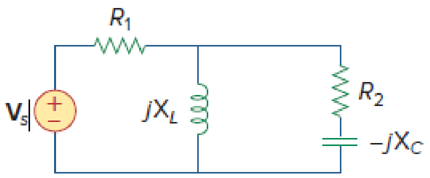
Chapter 11, Problem 4P

Problem

Using Fig. 11.36, design a problem to help other students better understand instantaneous and average power.

Figure 11.36:



Step-by-step solution

Step 1 of 4

Refer to Figure 11.36 in the textbook for the actual circuit.

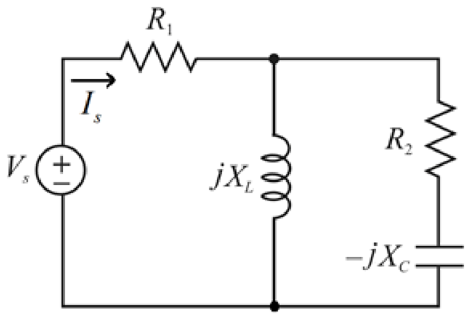


Figure 1

Step 2 of 4

Assume $v_s(t) = V_s \cos\omega t$ V

Determine the total impedance of the circuit.

R_2, $-jX_C$ are in series and in parallel with jX_L that is in series with R_1.

$$Z = \left[(R_2 - jX_C) \| jX_L\right] + R_1$$

$$= \left[\frac{(R_2 - jX_C)(jX_L)}{(R_2 - jX_C) + jX_L}\right] + R_1$$

$$= \left[\frac{X_C X_L + jR_2 X_L}{R_2 + j(X_L - X_C)}\right] + R_1$$

$$= \left[\frac{X_C X_L + jR_2 X_L + R_1 R_2 + jR_1(X_L - X_C)}{R_2 + j(X_L - X_C)}\right]$$

$$Z = \left[\frac{(R_1 R_2 + X_C X_L) + j\left((R_2 + R_1)X_L - R_1 X_C\right)}{R_2 + j(X_L - X_C)}\right]$$

Step 3 of 4

Recall the ohm's law,

$$R = \frac{V}{I}$$

Source current $I_s = \frac{V_s}{Z}$

$$I_s = \frac{V_s}{\left[\dfrac{(R_1R_2 + X_CX_L) + j((R_2+R_1)X_L - R_1X_C)}{R_2 + j(X_L - X_C)}\right]}$$

$$= \frac{R_2 + j(X_L - X_C)}{(R_1R_2 + X_CX_L) + j((R_2+R_1)X_L - R_1X_C)}V_s$$

$$I_s = \frac{\sqrt{R_2^2 + (X_L - X_C)^2}\angle\left(\tan^{-1}\left(\dfrac{X_L - X_C}{R_2}\right)\right)}{\sqrt{(R_1R_2 + X_CX_L)^2 + ((R_2+R_1)X_L - R_1X_C)^2}\angle\left(\tan^{-1}\left(\dfrac{(R_2+R_1)X_L - R_1X_C}{R_1R_2 + X_CX_L}\right)\right)}V_s$$

$$I_s = V_s\sqrt{\frac{R_2^2 + (X_L - X_C)^2}{(R_1R_2 + X_CX_L)^2 + ((R_2+R_1)X_L - R_1X_C)^2}}\angle\left(\begin{array}{l}\tan^{-1}\left(\dfrac{X_L - X_C}{R_2}\right) \\ \quad -\tan^{-1}\left(\dfrac{(R_2+R_1)X_L - R_1X_C}{R_1R_2 + X_CX_L}\right)\end{array}\right)$$

$$i_s(t) = K\cos\left[\omega t + \left(\begin{array}{l}\tan^{-1}\left(\dfrac{X_L - X_C}{R_2}\right) \\ \quad -\tan^{-1}\left(\dfrac{(R_2+R_1)X_L - R_1X_C}{R_1R_2 + X_CX_L}\right)\end{array}\right)\right]$$

Here, $K = V_s\sqrt{\dfrac{R_2^2 + (X_L - X_C)^2}{(R_1R_2 + X_CX_L)^2 + ((R_2+R_1)X_L - R_1X_C)^2}}$

Step 4 of 4

The average power delivered by the source is,

$$P=\frac{1}{2}V_sI_s\cos\left(\theta_v-\theta_i\right)$$

$$P=\frac{V_s^2}{2}\sqrt{\frac{R_2^2+\left(X_L-X_C\right)^2}{\left(R_1R_2+X_CX_L\right)^2+\left(\left(R_2+R_1\right)X_L-R_1X_C\right)^2}}\cos\left(\begin{array}{l}0-\tan^{-1}\left(\dfrac{X_L-X_C}{R_2}\right)\\ +\tan^{-1}\left(\dfrac{\left(R_2+R_1\right)X_L-R_1X_C}{R_1R_2+X_CX_L}\right)\end{array}\right)$$

Hence, the average power delivered by the source is,

$$\boxed{P=\frac{V_s^2}{2}\sqrt{\frac{R_2^2+\left(X_L-X_C\right)^2}{\left(R_1R_2+X_CX_L\right)^2+\left(\left(R_2+R_1\right)X_L-R_1X_C\right)^2}}\cos\left(\begin{array}{l}\tan^{-1}\left(\dfrac{\left(R_2+R_1\right)X_L-R_1X_C}{R_1R_2+X_CX_L}\right)\\ -\tan^{-1}\left(\dfrac{X_L-X_C}{R_2}\right)\end{array}\right)\text{W}}$$

The instantaneous power is given by,

$$p(t)=v_s(t)i_s(t)$$

$$p(t)=\left(V_s\cos\omega t\right)\left(K\cos\left[\omega t+\left(\begin{array}{l}\tan^{-1}\left(\dfrac{X_L-X_C}{R_2}\right)\\ -\tan^{-1}\left(\dfrac{\left(R_2+R_1\right)X_L-R_1X_C}{R_1R_2+X_CX_L}\right)\end{array}\right)\right]\right)$$

$$\boxed{p(t)=KV_s\cos\omega t.\cos\left[\omega t+\left(\begin{array}{l}\tan^{-1}\left(\dfrac{X_L-X_C}{R_2}\right)\\ -\tan^{-1}\left(\dfrac{\left(R_2+R_1\right)X_L-R_1X_C}{R_1R_2+X_CX_L}\right)\end{array}\right)\right]\text{W}}$$

Here, $K=V_s\sqrt{\dfrac{R_2^2+\left(X_L-X_C\right)^2}{\left(R_1R_2+X_CX_L\right)^2+\left(\left(R_2+R_1\right)X_L-R_1X_C\right)^2}}$